Total Derivative and chain rule:-

$u = f(x, y)$, $x = x(t)$, $y = y(t)$ then the composite function $u = f[x(t), y(t)]$ then

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\frac{dx}{dt} + \frac{\partial u}{\partial y}\frac{dy}{dt} \quad \text{——(1)}$$

$\frac{du}{dt}$ is called total differential co-efficient or total derivative.

→ Differential form

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy \quad \text{——(2)}$$

du is called the total differential of u. or Exact differential of u.

- If we put $t = x$ in (1),

$$\frac{du}{dx} = \frac{\partial u}{\partial x}\frac{dx}{dx} + \frac{\partial u}{\partial y}\frac{dy}{dx}.$$

i.e $\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx}$.

- $\frac{du}{dx}$ is called the total differential co-eff. of u w.r. to x

* Example:-

1. The time rate Q of flow through a cylindrical tube (such as a wind pipe) with radius r and length l is given by $Q = \frac{\pi p r^4}{8ln}$.

Scanned with CamScanner

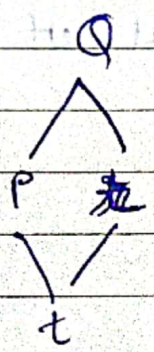
Where η is the viscosity of the fluid and p is difference in pressure at the two ends of the tube. Suppose the length of the tube remains constant, while the radius increases at the rate of $\frac{1}{10}$ and the pressure decreases at the rate of $\frac{1}{5}$. Find the rate of change of Q w.r. to time.

→

We have to find $\frac{dQ}{dt}$.

Here, $Q = \frac{\pi p r^4}{8l\eta}$, l = constant, $\frac{dr}{dt} = \frac{1}{10}$,

$\frac{dp}{dt} = \frac{-1}{5}$.



$$\frac{dQ}{dt} = \frac{\partial Q}{\partial p}\frac{dp}{dt} + \frac{\partial Q}{\partial r}\frac{dr}{dt}$$

$$= \frac{\pi r^4}{8l\eta}\left(\frac{-1}{5}\right) + \frac{4\pi p r^3}{8l\eta}\left(\frac{1}{10}\right)$$

$$= \frac{\pi r^3}{40l\eta}(2p - r)$$

$$\frac{dQ}{dt} = \frac{\pi r^3}{40l\eta}(2p - r).$$

* Differentiation of Implicit Functions:-

If $f(x,y) = 0$ represents an implicit func Suppose, $u = f(x,y)$ and $y = g(x)$ then the appropriate diagram is as follows.

Scanned with CamScanner

$f = x^y + y^x$

$\frac{\partial f}{\partial x} = y x^{y-1} + y^x \log y$

$\frac{\partial f}{\partial y} = x^y \log x + x y^{x-1}$

$\therefore \frac{dy}{dx} = -\left(\frac{y x^{y-1} + y^x \log y}{x^y \log x + x y^{x-1}} \right)$.

Partial Differentiation of Implicit Functions:

$F(x, y, z) = 0$ represents an implicit function, We want to determine $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

- We shall assume that $z = f(x, y)$.

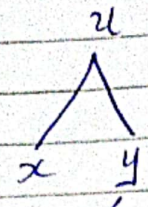
- Differentiating $F(x, y, z) = 0$ partially w.r. to x and y resp.

$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0 \quad \therefore \frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}}$.

$\frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial y} = 0 \quad \therefore \frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}}$, provided $\frac{\partial F}{\partial z} \neq 0$.

Consider x as the dependent variable & y, z as an independent variable,
i.e. $x = f(y, z)$

Scanned with CamScanner



$$\therefore \frac{dy}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx}$$

But $\frac{dy}{dx} = 0$ since, $u = f(x, y) = 0$

$$\therefore \frac{\partial y}{\partial x} + \frac{\partial y}{\partial y}\frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{\frac{\partial y}{\partial x}}{\frac{\partial y}{\partial y}} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{f_x}{f_y}$$

* Examples:-

1. Find $\frac{dy}{dx}$ when $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

→

Let, $f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{2ax + 2hy + 2g}{2hx + 2by + 2f}$$

$$\therefore \frac{dy}{dx} = -\left(\frac{2ax + 2hy + 2g}{2hx + 2by + 2f}\right)$$

2. Find $\frac{dy}{dx}$, if $x^y + y^x = a^b$

→

Let, $f(x, y) = x^y + y^x = a^b = \text{constant}$.

$$\frac{dy}{dx} = -\frac{f_x}{f_y}$$

Scanned with CamScanner

$$\frac{\partial x}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial x}} \quad \& \quad \frac{\partial x}{\partial z} = -\frac{\frac{\partial F}{\partial z}}{\frac{\partial F}{\partial x}}.$$

Ily, $\frac{\partial y}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}$, $\frac{\partial y}{\partial z} = -\frac{\frac{\partial F}{\partial z}}{\frac{\partial F}{\partial y}}$.

* Examples:-

1. Find the total differential co.efficient of x^2y w.r.to x when x, y are connected by $x^2 + xy + y^2 = 1$.

→

Let, $u = x^2y$, then the total differential co.eff. of u w.r. to x is

$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\frac{dy}{dx}$$

$$= 2xy + x^2\frac{dy}{dx}.$$

From the implicit relation,
$f = x^2 + xy + y^2 = 1$,

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{2x+y}{x+2y}.$$

So,

$$\frac{du}{dx} = 2xy + x^2\frac{dy}{dx} = 2xy + x^2\left(-\frac{2x+y}{x+2y}\right)$$

$$= 2xy - \frac{x^2(2x+y)}{x+2y}.$$

Scanned with CamScanner

2. Find $\frac{dy}{dx}$ when $y^{x^y} = \sin x$.

→

$$y^{x^y} = \sin x.$$

Taking log both the sides,

$$x^y \log y = \log(\sin x)$$

Let, $f(x,y) = x^y \log y - \log(\sin x)$ — (1)

Diff. (1) w.r.to x,

$$f_x = y x^{y-1} \log y - \frac{1}{\sin x} \cos x$$

$$= y x^{y-1} \log y - \cot x. \quad \text{— (2)}$$

Diff. (1) w.r.to y,

$$f_y = x^y \log x \log y + x^y \cdot \frac{1}{y} \quad \text{— (3)}$$

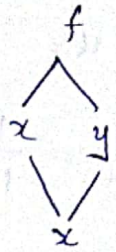
$$\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{(y x^{y-1} \log y - \cot x)}{x^y (\log x \cdot \log y + \frac{1}{y})}$$

Scanned with CamScanner

Implicit differentiation:

Any function of the type $f(x, y) = c$ is called an implicit function, where y is a function of x and c is a constant.

- If $f(x, y) = c$ then $\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$.



e.g. $y \log(\cos x) = x \log(\sin y)$

$(\cos x)^y = (\cos y)^x$

classroom problems:

1) If $x^3 + y^3 = 3axy$, find $\frac{dy}{dx}$.

$\frac{x^3+y^3}{xy} = 3a \Rightarrow f(x, y) = c$

$f(x, y) = x^3 + y^3 - 3axy$

$\frac{dy}{dx} = -\frac{df/dx}{df/dy} = -\frac{3x^2 - 3ay}{3y^2 - 3ax} = -\frac{-(ay - x^2)}{y^2 - ax} = \frac{ay - x^2}{y^2 - ax}$

2) If $x^x + y^y = c$, find $\frac{dy}{dx}$

$f(x, y) = x^x + y^x - c$

$\frac{dy}{dx} = -\frac{df/dx}{df/dy} = -\frac{3x^2}{3y^2} =$

3) If $x^y + y^x = c$, find $\frac{dy}{dx}$.

Let, $f(x, y) = x^y + y^x - c$

$\frac{dy}{dx} = -\frac{df/dx}{df/dy} = -\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$

4) If $u = \tan^{-1}\left(\frac{x}{y}\right)$ where $x^2 + y^2 = a^2$, find $\frac{dy}{dx}$.

$u = \tan^{-1}\frac{x}{y}$.

$\frac{du}{dx} =$

$f(x, y) = x^2 + y^2 - a^2$

$\frac{dy}{dx} = -\frac{df/dx}{df/dy}$

Scanned with CamScanner

$$\therefore \frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

$$\frac{dy}{dx} = \frac{\partial u}{\partial x}\frac{dx}{dx} + \frac{\partial u}{\partial y}\frac{dy}{dx}$$

Now, $u = \tan^{-1} x/y$

$$\frac{\partial u}{\partial x} = \frac{1}{1+x^2/y^2}\cdot\frac{1}{y} = \frac{y}{x^2+y^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{1+x^2/y^2}\left(\frac{-x}{y^2}\right) = \frac{-x}{x^2+y^2}$$

$$\therefore \frac{dy}{dx} = \frac{y}{x^2+y^2} + \left(\frac{-x}{x^2+y^2}\right)\left(-\frac{x}{y}\right)$$

$$= \frac{y}{x^2+y^2} + \frac{x^2}{y(x^2+y^2)}$$

$$= \frac{y^2+x^2}{y(x^2+y^2)}$$

$$= \frac{1}{y}.$$

(4) If $f(x,y)=0$, $\phi(x,z)=0$, show that $\frac{\partial\phi}{\partial x}\cdot\frac{\partial f}{\partial y}\cdot\frac{dy}{dz} = \frac{\partial f}{\partial x}\cdot\frac{\partial\phi}{\partial z}$

Here, $f(x,y)=0$ and $\phi(x,z)=0$

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \quad \text{and} \quad \frac{dz}{dx} = -\frac{\frac{\partial\phi}{\partial x}}{\frac{\partial\phi}{\partial z}}.$$

$$\frac{\frac{dy}{dx}}{\frac{dz}{dx}} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \Big/ -\frac{\frac{\partial\phi}{\partial x}}{\frac{\partial\phi}{\partial z}}$$

Scanned with CamScanner

$$\therefore \frac{dy}{dz} = \frac{\partial f/\partial x}{\partial f/\partial y} \Big/ \frac{\partial \phi/\partial x}{\partial \phi/\partial z}$$

$$\therefore \frac{dy}{dz} = \frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \times \frac{\frac{\partial \phi}{\partial z}}{\frac{\partial \phi}{\partial x}}$$

$$\therefore \frac{\partial \phi}{\partial x} \cdot \frac{\partial f}{\partial y} \cdot \frac{dy}{dz} = \frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial z}$$

If $y \log(\cos x) = x \log(\sin y)$. find $\frac{dy}{dx}$

$f(x, y) = y \log(\cos x) - x \log(\sin y)$

$$\frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y}$$

$$= -\frac{y \cdot \frac{1}{\cos x}(-\sin x) - \log(\sin y)}{\log(\cos x) - x \cdot \frac{1}{\sin y}\cos y}$$

$$= -\frac{\{-y \tan x - \log \sin y\}}{\log(\cos x) - x \cdot \cot y}$$

$$= \frac{y \tan x + \log \sin y}{\log \cos x - x \cot y}$$

Scanned with CamScanner